自觉遵守考场纪律　如考试作弊　此答卷无效

姓名_____

学号_____

线　封　密

东　南　大　学　考　试　试　卷（　卷）

课程代码	BJSL0040	课程名称	离散数学	考试学期	2021-2022-3
适用专业	计算机类	考试形式	闭　卷	考试时长	120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

题目	一	二	三	四	五	六	七	八	总分
得分									
批阅人									

1. Choice questions （ 2 points for each question ）

(1) Which of the following sentences are propositions? (C)

A. This sentence is wrong.

B. x+2=10.

C. 5+7=10.

D. What time is it?

(2). Which of the following set statements is wrong? (D)

A. $A \cup (A - B) = A$.

B. If $A \oplus B = \varnothing$, then $A = B$.

C. $|A \cup B| \leq |A| + |B|$.

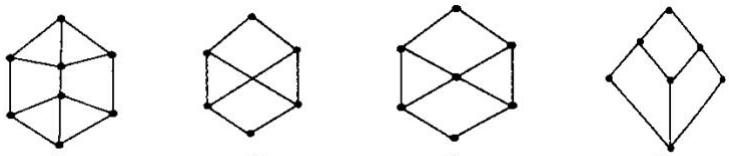
D. $P(A \times B) = P(A) \times P(A)$.

(3). Given the set A={1, 2, 3,4}，and the relation R={<1,1>, <2,2>, <2,3>, <3,2>, <3,3>,<4,4>}, R does not satsify（ D ）.

A. Reflexive.　　B. Symmetric.

C. Transitive.　　D. Antisymmetric.

(4). Which one of the following posets is NOT a lattice？(B)



A　　B　　C　　D

(5). For an undirected tree G with n vertices and m edges, which of the following statements is wrong？(D)

A. G is connected with m=n-1.
B. Any edge of G is a bridge.
C. There exists a unique simple path between any two vertices of G.
D. G can have a simple circuit.

(6). For a finite group G with order |G|=6, the order of subgroups of G cannot be (D)
A. 1 B. 2 C.3 D. 4

(7). Let H and K be two subgroups of group (G, +). Which one of the following sets associated with operation + is also a subgroup of G? (B)
A. H∪K B. H∩K C. K-H D. H⊕K

(8). For which pair of (m,n) does the complete bipartite graph $K_{m,n}$ have a Hamilton circuit? (D)
A. (1,1) B. (3,4) C. (4,8) D. (7,7)

(9). If an undirected tree G has 10 vertices of degree 3 and other vertices are leaves, how many leaves does G have? (D)
A. 9 B. 10 C.11 D. 12

(10). Suppose that G is a connected planar simple graph with 8 vertices, each of degree 3. How many regions does G has? (A)
A. 6 B. 7 C.8 D. 9

2. Completion （2 points for each question）

（1）Let P(x): x is a lion; Q(x): x is fierce; R(x): x drinks coffee. Express the following using quantifiers and P(x), Q(x), R(x)

a.) Some lions do not drink coffee. $\exists x(P(x) \wedge \neg R(x))$

b.) All lions are fierce. $\forall x(P(x) \rightarrow Q(x))$

(2) Let A ={a, b, c}, show the relation on set A which is symmetric and antisymmetric = {<a,a>,<b,b>,<c,c>}

(3) Let G is a forest consisting of K（K≥2）trees，then at least add K-1 edges to make G be a tree.

（4）Let <H，*>is a sub-group of group <G，*>. If |G|=n, |H|=m, the numeric relation

between m and n is _____m/n_____

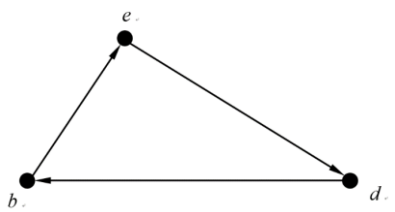
（5）Suppose that a connected planar graph has six vertices, each of degree four. The number of the regions in the plane divided by a planar representation of this graph is 8___

(6) Complete bipartite graph $K_{r,s}$ has an Euler circuit when r, s satisfy __r,s are even number_____.

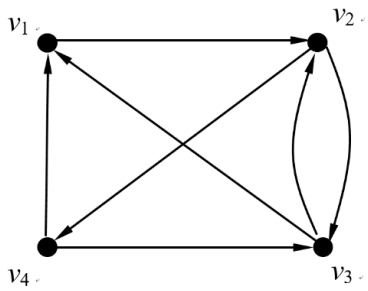
（7）Let S is no empty set and the algebra system (P(S)，∪，∩) is defined on the power set of S, the identity element with regard to the Union operation ∪ is ∅_____, the zero element with regard to the Union operation ∪ is ___S____

（8）The edge connectivity of complete graph K_n is $\lambda(K_n)$ =

3. **Let A={a, b, c, d, e, f}, the following graph depicted a relation R defined over Please give out the transitive closure t(R). (6 points)**



4. **A directed graph G = (V, E) is given below:**

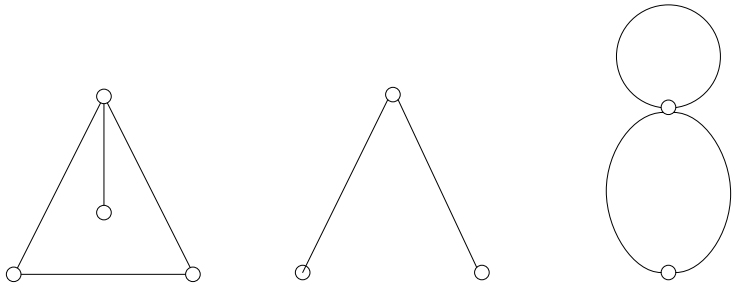


(1) How many paths from v4 to v2 whose length is equal to 2?
(2) **How many paths from v3 to v1 whose length is equal to 3?**
(3) How many circuits from v2 to v2 whose length is less than or equal to 3?

5. **Use necessary inference rules to prove that:**

$\{\forall x\,(P(x) \rightarrow (Q(y) \wedge R(x))), \exists x P(x)\} \vDash Q(y) \wedge \exists x(P(x) \wedge R(x))$

6. A planar graph G has k=3 connected components as shown in the following:



(1) **Find the relationship among the numbers of vertices, edges and regions of G. (2 points)**

(2) **If a planar graph G has k connected components, as well as n vertices, m edges and r regions, determine the relationship among these numbers and prove it. (10 points)**

7 Given an integer set B = {1,2,3,5,6,10,15,30}, ∀a, b ∈ B, a | b means a divides b, therefore | defines a partial relation on B.

(1) **Please draw the Hasse diagram of the partial relation |, and determine if it has the greatest and least elements. (6 points)**

(2) **Can you define three operations on the set B to explain why it is a Boolean algebra with these operations? (6 points)**

8 Let G be a group, define H={x| x ∈G , ∀ y ∈ G, xy=yx }.

(1) **Does H must be a sub-group of group G?**

(2) **Prove your conclusion.**